

CLEVELAND HOPKINS, OH, USA

WMO: 725240

Lat: 41.406N

Lon: 81.852W

Elev: 238

StdP: 98.50

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14820

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.8	-20.3	0.6	-14.7	-17.6	0.8	-12.0	12.8	-1.4	11.8	-2.4	4.7	230	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	32.1	23.1	30.7	22.4	29.2	21.7	24.5	29.7	23.7	28.5	22.9	27.5	4.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.1	27.4	22.1	17.3	26.6	21.3	16.5	25.8	75.2	30.0	71.8	28.4	68.7	27.6	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.4	8.5	-18.6	34.0	4.0	1.4	-21.4	35.0	-23.8	35.8	-26.0	36.5	-28.9	37.5		
(5)			-19.0	26.0	3.9	1.1	-21.8	26.8	-24.1	27.5	-26.3	28.1	-29.2	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-2.4	-1.0	3.3	9.8	15.8	21.0	23.0	22.3	18.8	12.5	6.2	0.8	(6)
(7)		DBStd	10.28	6.57	6.28	6.14	5.37	4.89	3.86	3.04	2.90	3.97	4.72	5.16	5.49	(7)
(8)		HDD10.0	1453	387	313	226	68	7	0	0	1	26	137	289		(8)
(9)		HDD18.3	3187	643	542	468	260	112	18	2	3	42	189	364	544	(9)
(10)		CDD10.0	1782	3	4	18	63	185	329	404	382	263	104	23	4	(10)
(11)		CDD18.3	474	0	0	1	5	32	98	148	127	54	9	0	0	(11)
(12)		CDH23.3	3747	0	0	6	51	277	788	1241	926	397	59	1	0	(12)
(13)		CDH26.7	1092	0	0	0	7	68	246	398	253	112	8	0	0	(13)
(14)	Wind	WSAvg	4.3	5.1	4.8	4.8	4.7	4.1	3.7	3.5	3.3	3.6	4.2	4.6	4.8	(14)
(15)	Precipitation	PrecAvg	937	57	56	75	82	91	89	88	87	89	66	81	76	(15)
(16)		PrecMax	1367	113	119	132	168	232	230	232	228	186	141	224	218	(16)
(17)		PrecMin	477	9	12	22	32	25	16	17	14	19	18	20	28	(17)
(18)		PrecStd	161	26	25	26	30	44	45	38	45	39	28	42	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	19.3	23.6	27.8	30.6	33.0	33.9	32.9	32.5	27.6	21.6	17.3	(19)
(20)		MCWB	13.4	13.9	15.8	18.1	21.0	22.8	24.7	24.1	22.6	20.2	15.3	13.8		(20)
(21)		2%	DB	13.5	14.4	19.7	24.7	28.6	31.3	32.0	31.1	29.7	25.0	18.6	14.3	(21)
(22)		MCWB	11.3	10.7	13.5	16.2	20.3	22.5	23.5	22.9	21.8	18.4	14.1	11.4		(22)
(23)		5%	DB	10.3	10.8	16.6	21.9	26.7	29.6	30.5	29.4	27.5	22.6	16.5	11.3	(23)
(24)		MCWB	7.9	7.6	12.1	14.7	19.3	21.8	22.7	22.2	20.6	17.0	12.6	9.0		(24)
(25)		10%	DB	6.7	7.7	13.3	19.1	24.3	27.7	28.9	27.9	25.5	20.2	14.3	8.4	(25)
(26)		MCWB	4.5	5.1	9.4	13.1	18.0	20.9	21.9	21.5	19.6	15.8	10.7	6.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	14.8	16.6	19.8	22.9	24.7	26.1	25.5	24.1	21.6	17.3	14.7	(27)
(28)		MCDWB	15.9	18.9	22.4	24.9	28.6	30.4	32.1	30.7	29.8	26.4	20.4	16.8		(28)
(29)		2%	WB	11.6	11.3	14.5	17.5	21.4	23.6	24.8	24.2	22.9	19.8	15.0	12.0	(29)
(30)		MCDWB	13.6	13.8	18.6	23.0	26.8	29.2	30.4	28.6	27.6	23.3	18.0	13.9		(30)
(31)		5%	WB	8.3	8.0	12.4	15.6	20.2	22.7	23.8	23.4	21.7	17.9	13.1	9.1	(31)
(32)		MCDWB	10.0	10.5	16.1	20.7	25.1	28.0	28.5	27.5	25.7	21.5	16.0	11.0		(32)
(33)		10%	WB	4.6	5.0	9.7	13.8	18.9	21.8	22.9	22.5	20.6	16.3	11.0	6.4	(33)
(34)		MCDWB	6.6	7.7	13.5	18.6	23.4	26.6	27.3	26.5	24.4	19.6	13.6	8.4		(34)
(35)	Mean Daily Temperature Range	MDBR	7.1	7.6	8.5	10.0	10.1	9.5	9.3	9.1	9.5	8.8	7.6	6.4		(35)
(36)		5% DB	MCDBR	10.2	11.8	13.3	13.9	12.5	11.3	10.9	10.8	11.9	11.7	10.5	9.5	(36)
(37)		MCWBR	8.9	9.4	8.8	8.6	6.7	5.3	4.9	5.0	5.7	6.7	7.7	8.0	(37)	
(38)		5% WB	MCDBR	10.0	11.2	12.2	12.6	11.0	10.2	9.4	9.1	9.4	9.9	9.7	9.5	(38)
(39)		MCWBR	9.2	9.7	8.9	8.8	6.7	5.4	4.7	4.8	5.1	6.5	8.0	8.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.335	0.357	0.395	0.431	0.449	0.447	0.433	0.385	0.367	0.326	0.309	(40)	
(41)		taud	2.403	2.279	2.297	2.278	2.213	2.208	2.234	2.260	2.370	2.399	2.483	2.469	(41)	
(42)		Ebn at Noon	845	865	887	876	849	831	828	829	849	821	816	812	(42)	
(43)		Edh at Noon	80	106	117	128	141	142	137	130	108	94	75	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.38	2.24	3.29	4.49	5.38	5.96	5.98	5.29	4.23	2.71	1.72	1.13	(44)	
(45)		RadStd	0.13	0.29	0.32	0.39	0.46	0.44	0.38	0.24	0.38	0.23	0.27	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) +0.56	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +148	(m) +87	(46)
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	N/A	(47)

Nomenclature: See separate page